

Slide 2

EARTH'S INTERNAL STRUCTURE

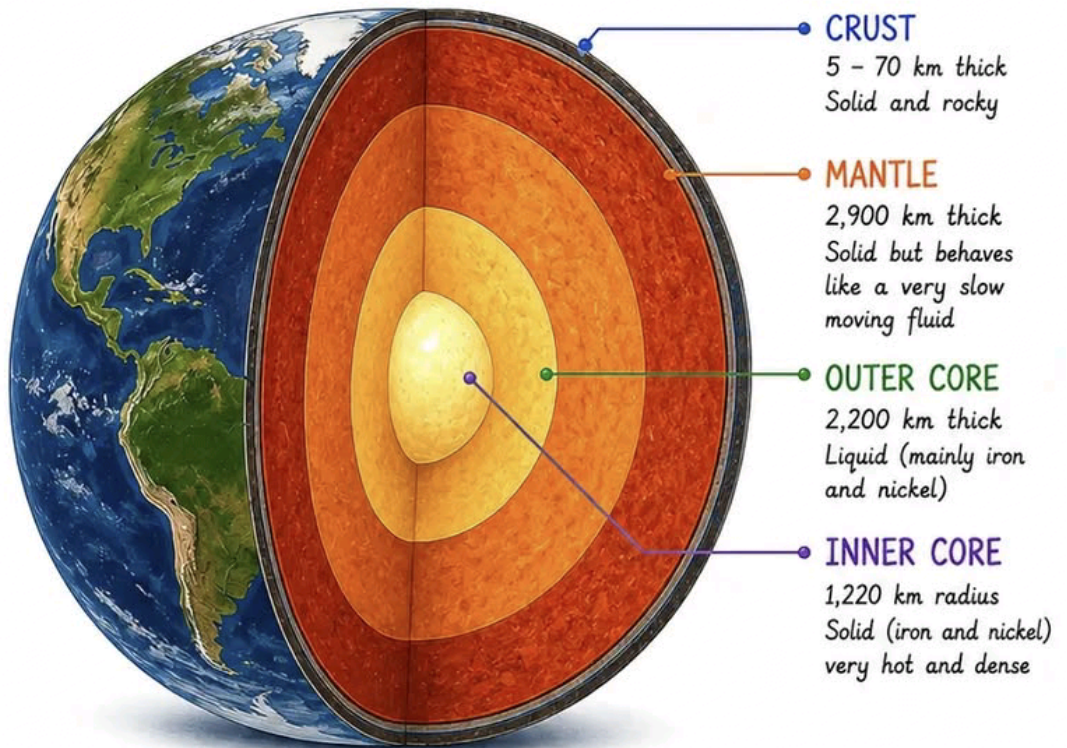
Layers of the Earth

OVERVIEW

- The Earth is made up of layers based on **composition**, **temperature** and physical properties.

KEY POINTS

- ★ Temperature and pressure **increase** with depth.
- ★ These layers work together to keep the Earth **active** and dynamic.
- ★ **Plate tectonics** occurs in the outermost solid layer.



LAYERS AT A GLANCE

LAYER	THICKNESS	STATE	COMPOSITION	TEMPERATURE (APPROX.)
Crust	5 - 70 km	Solid	Rocks (Silicates)	0°C - 700°C
Mantle	~ 2,900 km	Solid (plastic)	Silicate rocks	~ 700°C - 3,700°C
Outer Core	~ 2,200 km	Liquid	Iron, Nickel	~ 3,700°C - 5,000°C
Inner Core	~ 1,220 km (radius)	Solid	Iron, Nickel	~ 5,000°C - 6,000°C



Earth's internal structure controls many natural events like **earthquakes**, **volcanoes** and **mountain building**.



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WHAT ARE TECTONIC PLATES?

Brain
Yatra
Notes

Large pieces of Earth's lithosphere
that *move* and *interact*.

DEFINITION

- Tectonic plates are *huge*, *rigid* pieces of Earth's lithosphere (crust + uppermost mantle).
- They *move* slowly over the semi-fluid *asthenosphere*.

KEY FEATURES ★

- ➡ Plates vary in *size*.
- ➡ They have well-defined *boundaries*.
- ➡ They move a few *centimeters* per year.
- ➡ Their movement *shapes* Earth's surface.

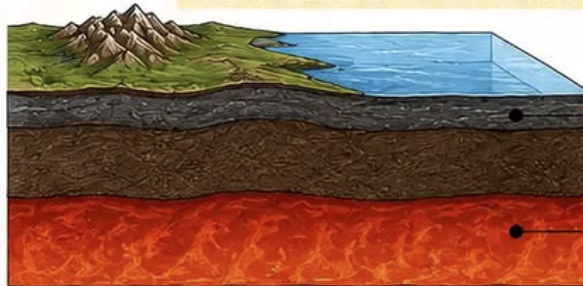


Plates are
rigid

They *move*
slowly

They interact
at their
boundaries

RELATIONSHIP WITH EARTH'S LAYERS



Lithosphere
(Crust + Uppermost Mantle)
Rigid and broken into plates.

Asthenosphere
Soft, semi-fluid layer.
Allows plates to move.



IMPORTANT POINT 💡

Tectonic plates are *not fixed*.
They are always moving and
changing Earth's surface.

Examples of what they cause:



Mountains



Volcanoes



Earthquakes



Tsunamis

Slide 5

MAJOR & MINOR TECTONIC PLATES

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Notes

Earth's lithosphere is divided into
7 major and several minor plates.

OVERVIEW

- The Earth's outer shell is broken into **plates**.
- There are **7 major** plates and many **minor** plates.
- Plates vary in size and move in **different directions**.

7 MAJOR PLATES

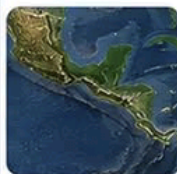
- 1 Pacific Plate
- 2 North American Plate
- 3 South American Plate
- 4 African Plate
- 5 Eurasian Plate
- 6 Antarctic Plate
- 7 Indo-Australian Plate



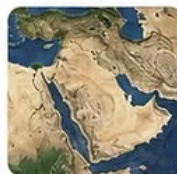
IMPORTANT FACTS

- ✓ The Pacific Plate is the **largest**.
- ✓ The Antarctic Plate surrounds **Antarctica**.
- ✓ The **Arabian Plate** is a small continental plate.
- ✓ Plates are always **moving**, not fixed.

SOME IMPORTANT MINOR PLATES



Caribbean Plate



Arabian Plate



Philippine Sea Plate



Nazca Plate



Anatolian Plate



These plates interact at their boundaries, causing **earthquakes**, **volcanoes**, mountain building and ocean trench formation.



Slide 1

PLATE TECTONICS

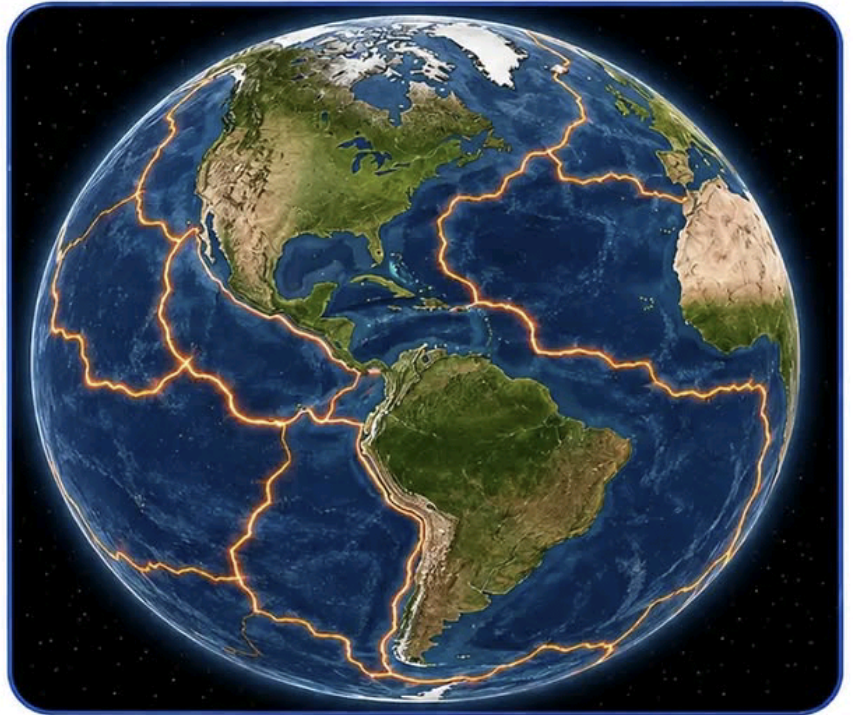


The Dynamic Earth



INTRODUCTION

- Earth's **outer shell** is broken into large pieces called **tectonic plates**.
- These plates are always **moving**, though very slowly.
- The movement of plates changes the **shape** of Earth's surface.



KEY POINTS



Earth is made of **moving** plates.



Plate movement causes **mountains**, **earthquakes**, **volcanoes** and **oceans**.



It is a slow process, but its effects are **powerful**.



It has been shaping our planet for **millions** of years.



"The Earth is not still, it is **dynamic**. Plate tectonics is the reason."



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WHY DO PLATES MOVE?

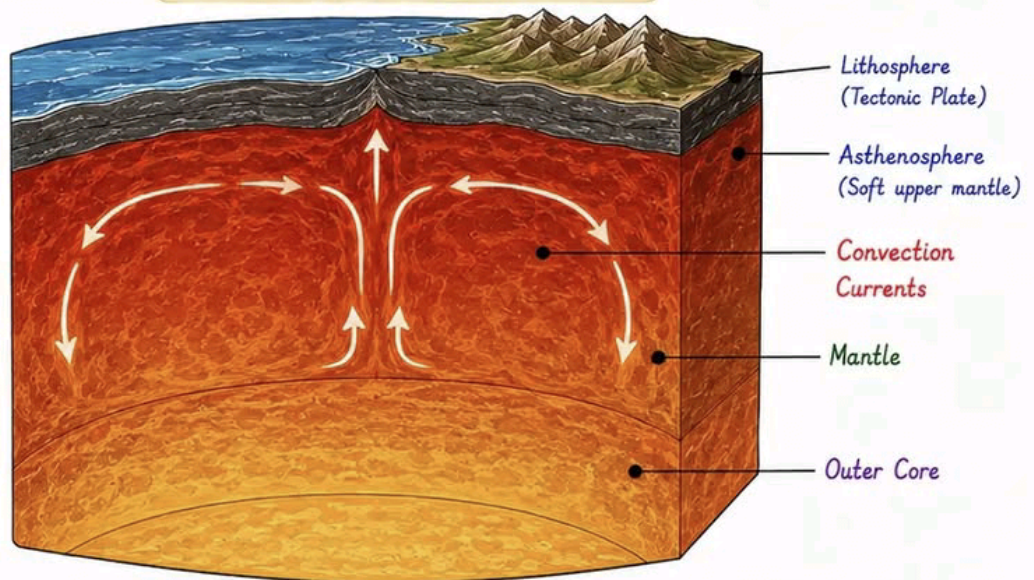
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Notes

Heat from inside the Earth causes movement in the **mantle**. This movement **drives** the plates.

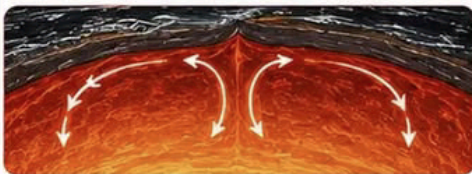
MAIN REASONS

- **Convection Currents**
Heat from Earth's core makes mantle material rise, spread out, cool, sink and flow back.
- **Ridge Push**
Raised ridges at mid-ocean spreading centers push plates away from the ridge due to gravity.
- **Slab Pull**
When an oceanic plate sinks into the mantle at subduction zones, it **pulls** the rest of the plate behind it.

CONVECTION CURRENTS IN THE MANTLE

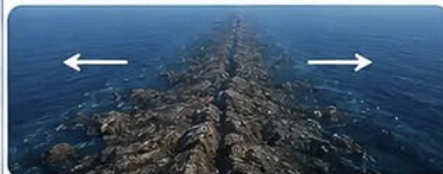


1. CONVECTION CURRENTS



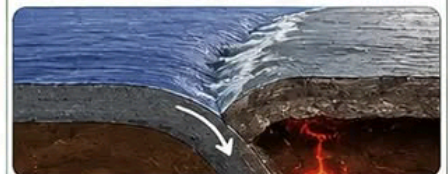
Heat from the core causes hot mantle material to **rise**, cool near the surface, sink down and flow again in a cycle. This slow movement drags the plates on top of it.

2. RIDGE PUSH



New crust formed at mid-ocean ridges is higher. Due to gravity, it slides down the slope and **pushes** the plates away from the ridge.

3. SLAB PULL



Cold, dense oceanic plate sinks into the mantle at subduction zones. Its weight **pulls** the rest of the plate behind it.



KEY POINT

Convection currents in the mantle are the main driving force, while **Ridge Push** and **Slab Pull** help in moving the plates.



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LITHOSPHERE & ASTHENOSPHERE

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The Strong Layer & The Flowing Layer

OVERVIEW

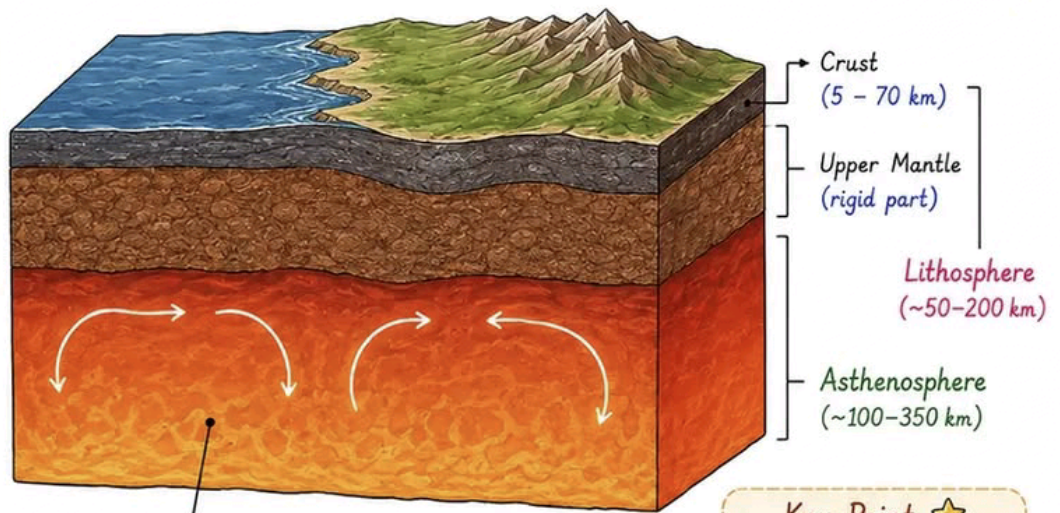
- The Earth's outer part is divided into two important layers of the **upper mantle**.
- These layers play a key role in **plate tectonics**.

LITHOSPHERE

- The outermost, **rigid** and **strong** layer.
- It includes the **crust** and the **uppermost mantle**.
- Broken into **tectonic plates** that move over the asthenosphere.
- Thickness: ~ 50 to 200 km

ASTHENOSPHERE

- The layer **below** the lithosphere.
- Hotter**, **soft** and **weak** zone of the upper mantle.
- It behaves like a **plastic** or **semi-molten** material.
- Thickness: ~ 100 to 350 km



Semi-molten material that flows slowly.

Key Point

Lithosphere is **strong** and **breaks**.
Asthenosphere is **weak** and **flows**.

LITHOSPHERE vs ASTHENOSPHERE

	Lithosphere	Asthenosphere
	Rigid and strong	Soft and weak
	Broken into tectonic plates	Allows plates to move
	Cooler	Hotter
	Thickness: ~ 50-200 km	Thickness: ~ 100-350 km



Lithosphere **rides** on the **asthenosphere**.



This movement causes **earthquakes**, **volcanoes** and **mountain building**.



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VOLCANOES & PLATE TECTONICS

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Notes



Volcanoes are formed when molten rock from inside the Earth reaches the surface. Most volcanoes are found at **plate boundaries**.

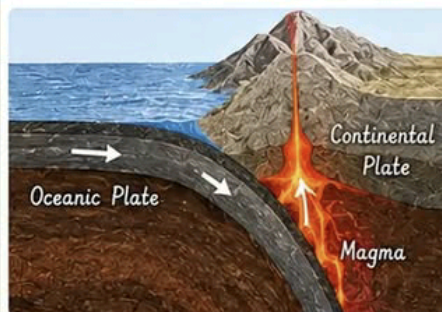
HOW VOLCANOES FORM?

- 1 At **convergent boundaries**, one plate subducts under another.
- 2 As the plate sinks, it **melts** in the mantle.
- 3 The **magma** rises through cracks and weak zones.
- 4 When magma reaches the surface, it erupts as **lava**, **ash** and **gases**.



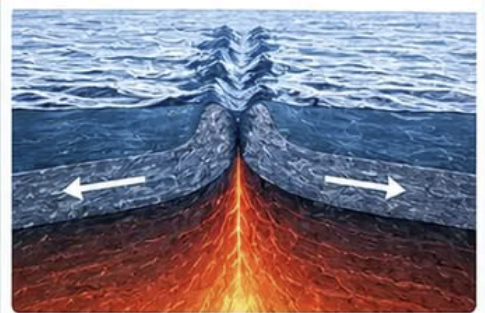
Where Volcanoes are Found?

CONVERGENT BOUNDARY (SUBDUCTION ZONE)



Example: Andes Mountains

DIVERGENT BOUNDARY (MID-OCEAN RIDGE)



Example: Mid-Atlantic Ridge

TYPES OF VOLCANOES

SHIELD VOLCANO



Broad, gentle slopes.
Built by fluid lava.
Example: Mauna Loa (Hawaii)

COMPOSITE VOLCANO



Steep sides, formed by layers of lava and ash.
Example: Mount Fuji (Japan)

CINDER CONE



Small, steep-sided cones made of ash and rock fragments.
Example: Parícutin (Mexico)



KEY POINTS

- Most volcanoes occur at **plate boundaries**.
- Subduction zones create **explosive** volcanoes.
- Divergent boundaries create **gentle lava flows**.
- Volcanoes shape landforms and affect the climate.



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MOUNTAINS & PLATE TECTONICS

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Notes

Mountains are formed by the movement and collision of **tectonic plates**.

HOW MOUNTAINS FORM?

- 1 When two continental plates **collide**, the crust crumples and folds.
- 2 The rocks are pushed upward forming mountain ranges.
- 3 This process is called **orogenesis**.
- 4 Examples: Himalayas, Andes, Alps.

Formation of Fold Mountains (Collision of Plates)



Example: The Himalayas (India-Eurasia Plate Collision)

TYPES OF MOUNTAINS

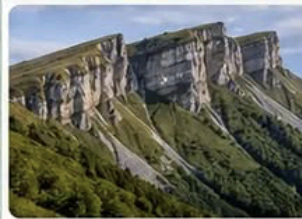
- **Fold Mountains**
Formed by the collision of continental plates.
- **Block Mountains**
Formed by the faulting of the Earth's crust.
- **Volcanic Mountains**
Formed by the eruption of magma.

FOLD MOUNTAINS



Formed by collision of plates. **Example:** Himalayas

BLOCK MOUNTAINS



Formed by large faults in the Earth's crust. **Example:** Vosges (France)

VOLCANIC MOUNTAINS



Formed by volcanic eruptions. **Example:** Mount Fuji (Japan)



KEY POINT

Mountains are not permanent. Over millions of years, they are also worn down by **weathering** and **erosion**.



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TYPES OF PLATE BOUNDARIES

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Notes

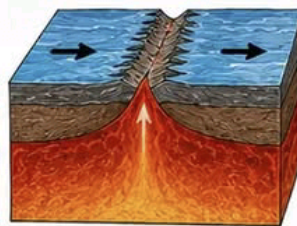
Where plates meet, different kinds of **boundaries** are formed.

THERE ARE 3 MAIN TYPES

- 1 **Divergent Boundary**
Plates move **away** from each other. Magma rises up and new crust is formed.
- 2 **Convergent Boundary**
Plates move **toward** each other. One plate may go under the other (**subduction**) or both may collide to form **mountains**.
- 3 **Transform Boundary**
Plates **slide past** each other horizontally along a **fault**.

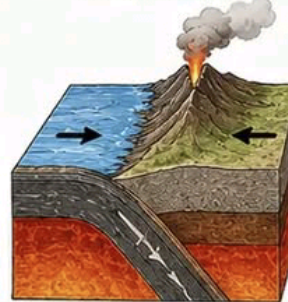
DIAGRAM OF PLATE BOUNDARIES

DIVERGENT



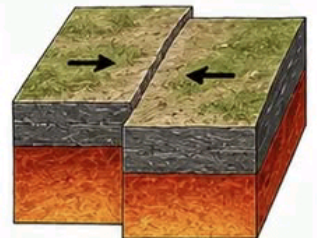
Plates move apart.
Magma rises up and new crust forms.

CONVERGENT



Plates move together.
One plate goes under the other or both collide.

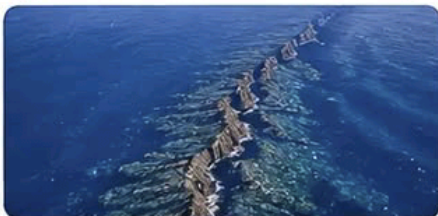
TRANSFORM



Plates slide past each other sideways along a fault.

REAL WORLD EXAMPLES

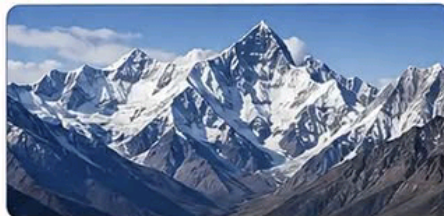
1 DIVERGENT – Mid-Atlantic Ridge



Location: **Mid-Atlantic Ocean**

New oceanic crust is formed as the North American Plate and Eurasian Plate move apart.

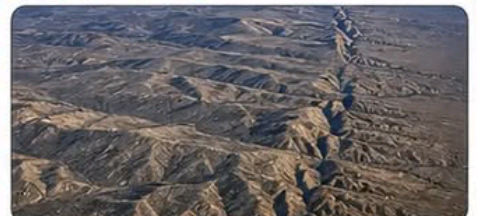
2 CONVERGENT – Himalayas



Location: **India (Eurasian Plate & Indian Plate)**

The Indian Plate **collides** with the Eurasian Plate, forming the **Himalayas**.

3 TRANSFORM – San Andreas Fault



Location: **California, USA**

The Pacific Plate and North American Plate **slide past** each other along the San Andreas **Fault**.



KEY POINT →

These boundaries are responsible for **earthquakes**, **volcanoes**, **mountain building**, and the creation of **new crust**.



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EARTHQUAKES & PLATE MOVEMENT

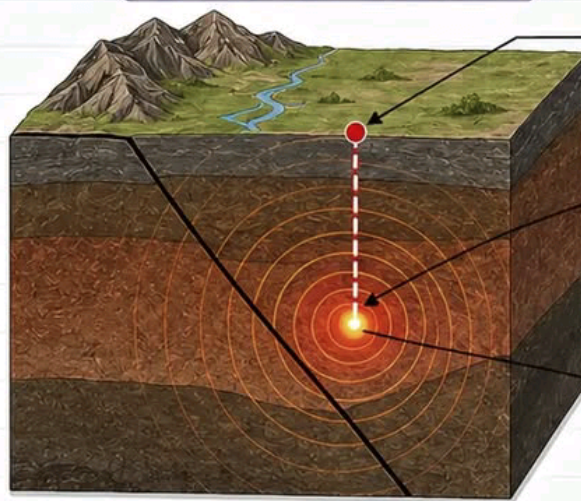
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Notes

Earthquakes occur when **stress** builds up in rocks due to **plate movement** and is released suddenly.

HOW EARTHQUAKES HAPPEN?

- 1 Tectonic plates are always moving.
- 2 Sometimes, plates get stuck at **faults**.
- 3 Stress builds up in rocks over time.
- 4 When the stress is too much, rocks break or **slip** suddenly.
- 5 Energy is released in the form of **seismic waves**, causing an earthquake.

PARTS OF AN EARTHQUAKE



Epicenter

Point on the Earth's surface above the focus.

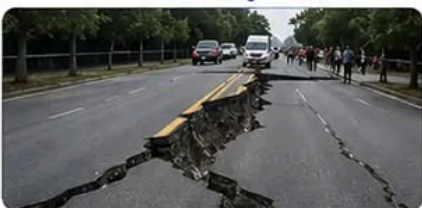
Focus (Hypocenter)

Actual underground point where the earthquake starts.

Fault

Crack in rocks along which movement occurs.

P-Waves (Primary Waves)



Fastest waves, travel through solids, liquids and gases.

TYPES OF SEISMIC WAVES

S-Waves (Secondary Waves)



Slower than P-waves, travel only through solids.

Surface Waves (Love & Rayleigh Waves)



Slowest but most destructive, cause the most damage.



KEY POINT



Earthquakes are nature's way of releasing built-up **stress** due to the movement of **tectonic plates**.



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PLATE TECTONICS

REVISION SHEET

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Notes



The Earth is *dynamic*. Its surface is always changing because of the movement of *tectonic plates*.



KEY POINTS

- Earth's outer layer (lithosphere) is broken into large and small tectonic plates.
- Plates move due to *convection currents* in the mantle.
- These movements cause *earthquakes*, *volcanoes*, *mountain building*, and *ocean floor changes*.

TYPES OF PLATE BOUNDARIES

1 Divergent Boundary

Plates move apart.
New crust forms.
(e.g., Mid-Atlantic Ridge)



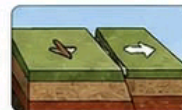
2 Convergent Boundary

Plates move toward each other.
Subduction or collision occurs.
(e.g., Himalayas, Japan Trench)



3 Transform Boundary

Plates slide past each other.
(e.g., San Andreas Fault)



MAJOR EFFECTS



Mountain Building
(e.g., Himalayas, Andes)



Volcanic Activity
(e.g., Mt. Fuji, Kilauea)



Earthquakes
(e.g., Japan, Nepal)



Ocean Floor Changes
(e.g., Mid-Ocean Ridges, Ocean Trenches)

IMPORTANT TERMS

- **Lithosphere:** Rigid outer layer (crust + uppermost mantle).
- **Asthenosphere:** Soft, weak layer below lithosphere.
- **Convection Current:** Heat-driven movement in the mantle.
- **Subduction:** Sinking of one plate beneath another.
- **Mid-Ocean Ridge:** Underwater mountain range where new crust is formed.

PLATE TECTONICS IN ACTION



Himalayas were formed by the collision of the Indian Plate and the Eurasian Plate.

REMEMBER

- ✓ Plates are always moving.
- ✓ Slow movement, big changes.
- ✓ Natural events are linked to plate movements.
- ✓ Understanding today, preparing for tomorrow.

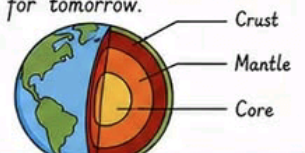


Plate tectonics is the key to understanding our Earth's *past*, *present* and *future*.



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REAL-WORLD EXAMPLES OF PLATE TECTONICS

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Notes



Plate movements shape our planet - *Mountains*, *Earthquakes*, *Volcanoes*, and *Oceans* are all linked to plate tectonics.



1. HIMALAYAS - MOUNTAIN BUILDING



- Formed by the collision of the Indian Plate with the Eurasian Plate.
- This collision continues today.

2. JAPAN - EARTHQUAKES



- Japan lies at convergent boundaries of the Pacific, Philippine, and Eurasian Plates.
- Frequent earthquakes occur here.

3. ICELAND - VOLCANIC ACTIVITY



- Iceland sits on the Mid-Atlantic Ridge (divergent boundary).
- Magma rises up, causing volcanic eruptions.

4. ANDES MOUNTAINS - SUBDUCTION

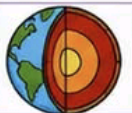


- The Nazca Plate subducts under the South American Plate.
- This forms the Andes Mountains and causes earthquakes & volcanoes.



KEY TAKEAWAY

Plate tectonics explains many *natural events* and *landforms*.
Understanding it helps us *predict* and *prepare* for natural disasters.



REBUILDING & LIVING SAFELY

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Notes



We cannot stop earthquakes,
but we can be **prepared**, stay **safe**
and **rebuild** better.



WHAT CAN WE DO?



Follow **building** codes and use earthquake-resistant designs.



During an earthquake, Drop, Cover and Hold On.



Keep an **emergency** kit ready with food, water, torch, first aid and important documents.

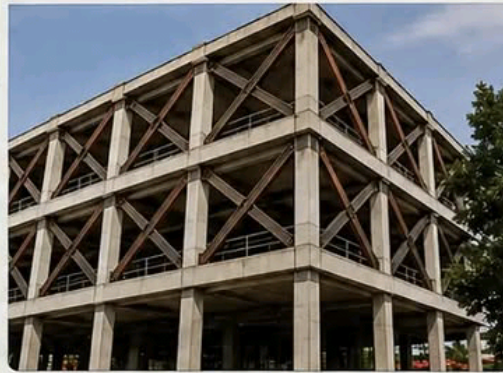


Know the **safe exit** routes at home, school and workplace.



Be **aware** and **help** others. Community preparedness saves lives.

EARTHQUAKE-RESISTANT BUILDING



STAY SAFE, STAY PREPARED



REBUILDING BETTER



REMEMBER



Preparation today
can protect our
lives tomorrow.

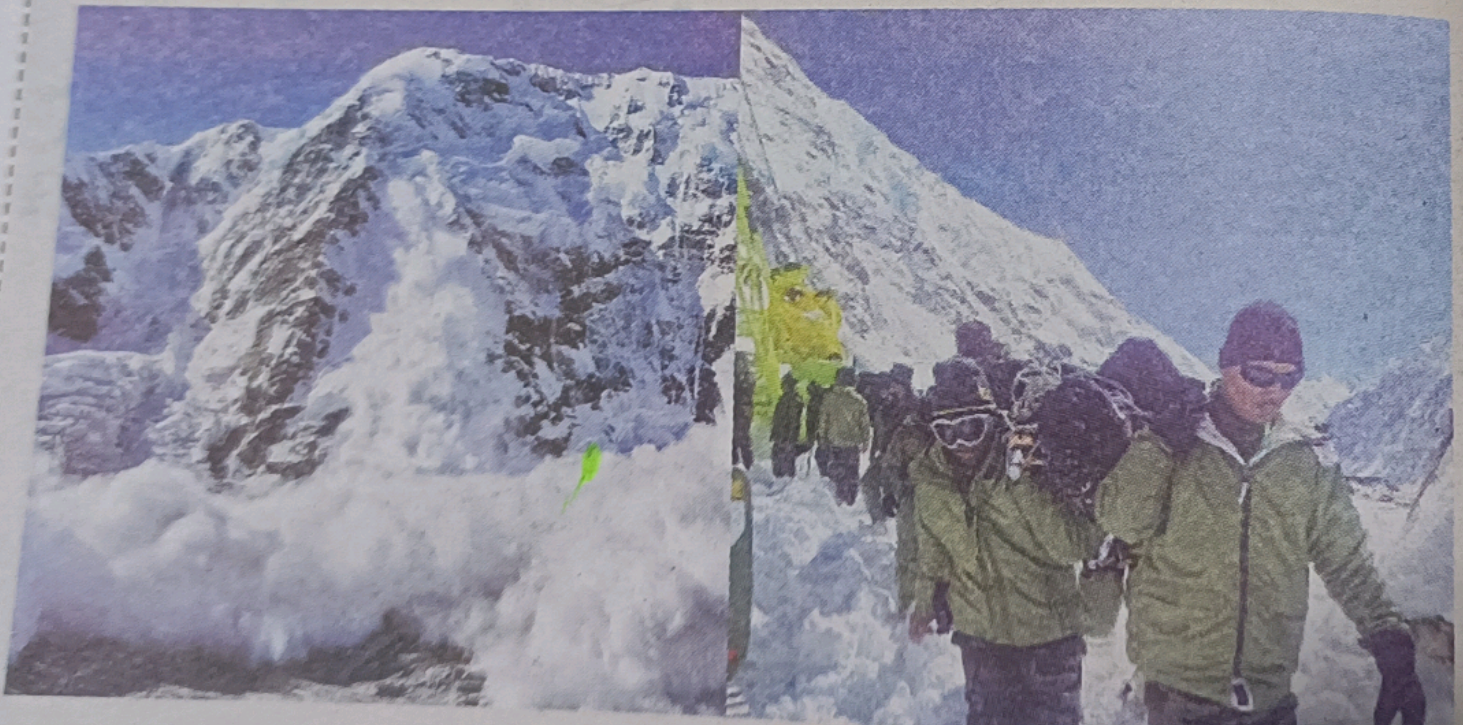


Earthquakes are a **part of nature**,
but **safety and awareness** are
in our hands.



Avalanches

Avalanches are caused by the sudden instability of snow on steep mountain slopes. Heavy snowfall within a short period adds extra weight to the snowpack, making it unstable, especially when it rests on weak or loosely bonded layers of snow. A sudden rise in temperature can cause partial melting, reducing the friction that holds the snow together. Strong winds may also pile up snow unevenly, creating fragile layers. Natural disturbances, such as earthquakes and vibrations, as well as human activities like skiing, trekking, or construction in mountainous areas, can trigger avalanches by disturbing the balance of the snow-covered slopes.



GLOFs

Glacial Lake Outburst Floods (GLOFs) are caused by the sudden release of large volumes of water from glacial lakes due to natural and climatic factors. Rapid melting of glaciers because of rising temperatures increases the size and water level of glacial lakes, putting pressure on their natural dams made of ice or loose moraines. Heavy rainfall or intense snowfall can add excess water to these lakes. Earthquakes, avalanches, or landslides may strike the lake or weaken the dam, leading to its sudden collapse. As a result, the stored water is released abruptly, causing destructive floods in downstream areas.

